

Association Analysis Using Linear Regression: Predicting Calorie Expenditure During Exercise

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I. INTRODUCTION

Predicting calorie expenditure during exercise is important for evaluating physical activity and health outcomes. This study applies single and multivariable linear regression to model calorie burn using physiological and demographic features from a large exercise dataset.

II. METHODOLOGY

A. Dataset Selection and Description

The dataset selected for this analysis is the "Calories Burnt Prediction" dataset from Kaggle. This dataset was chosen because it satisfies all specified requirements for multivariable linear regression analysis:

- **Number of predictors:** The dataset contains 7 predictors after preprocessing, exceeding the minimum of 3.
- **Sample size:** With 15,000 observations and 7 predictors, the condition $n \geq p + 5$ is satisfied ($15,000 \geq 12$).
- **Response variable:** Continuous numerical variable suitable for linear regression.

B. Variable Classification

Quantitative Variables:

- Age: Continuous numerical (years)
- Height: Continuous numerical (cm)
- Weight: Continuous numerical (kg)
- Duration: Continuous numerical (minutes)
- Heart_Rate: Continuous numerical (bpm)
- Body_Temp: Continuous numerical ($^{\circ}\text{C}$)
- Calories: Continuous numerical response (kcal)

Categorical Variables:

- Gender: Binary categorical variable (Female/Male)

C. Outlier Detection and Removal

Outliers were detected using the Interquartile Range (IQR) method, a robust statistical technique that identifies extreme values without assuming normal distribution. The implementation steps were:

- 1) Calculated the first quartile (Q1) and third quartile (Q3) for each quantitative variable.
- 2) Computed the interquartile range: $IQR = Q3 - Q1$.
- 3) Defined outlier boundaries:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

- 4) Flagged and removed observations falling outside these boundaries (excluding the target variable).

D. Descriptive Statistics Calculation

Comprehensive descriptive statistics were calculated for all quantitative variables:

Measures of Central Tendency:

- **Mean:** Arithmetic average ($\bar{x} = \frac{\sum x_i}{n}$).
- **Median:** Middle value when sorted.
- **Mode:** Most frequently occurring value.

Measures of Dispersion:

- **Standard Deviation:** Spread around mean ($s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$).
- **Variance:** Average of squared deviations (s^2).
- **Range:** Difference between maximum and minimum.
- **IQR:** Difference between 75th and 25th percentiles.

Distribution Shape:

- **Skewness:** Measure of asymmetry.
- **Kurtosis:** Measure of tail heaviness.

E. Feature Standardization

To ensure numerical stability and comparability between predictors with different units (e.g., Age vs. Heart Rate), quantitative features were standardized using Z-Score Normalization. For each feature x , the standardized value z is:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

F. Categorical Variable Encoding

One-Hot Encoding was applied to convert categorical variables into binary indicators. To prevent perfect multicollinearity ("Dummy Variable Trap"), we utilized $k - 1$ encoding. For a variable with k levels, only $k - 1$ binary columns were generated. This ensures the design matrix remains full rank, satisfying OLS assumptions.

G. Statistical Analysis Framework

To verify regression assumptions, an automated framework combined visual diagnostics, formal normality testing, and correlation analysis.

1) **Distributional Assessment:** Univariate distributions were examined using histograms (shape), boxplots (spread/outliers), and Q-Q plots (tail deviations). Visual findings were supported by formal hypothesis testing ($\alpha = 0.05$):

- H_0 : The feature follows a Normal distribution.
- H_1 : The feature does not follow a Normal distribution.

Given the large sample size ($N > 5000$), two tests were applied: Anderson-Darling (sensitive to tails) and D'Agostino-Pearson (skewness/kurtosis).

2) *Correlation Analysis*: Pearson correlation coefficients were computed between the response variable (*Calories*) and each predictor to quantify linear associations and identify potential multicollinearity.

H. Regression Analysis

1) *Single-Feature Linear Regression*: To model the relationship between individual physiological predictors and calorie expenditure, a simple linear regression model was employed, which minimizes the squared residuals between observed and predicted values. The model is defined as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

To demonstrate a fundamental understanding of regression mechanics, the regression coefficients were computed manually using closed-form solutions based on sample covariance and variance. Specifically, the sample covariance (S_{xy}) and sample variance (S_{xx}) were calculated as:

$$S_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1} \quad S_{xx} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

These quantities were then used to derive the slope and intercept:

$$b_1 = \frac{S_{xy}}{S_{xx}}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

To validate the correctness of the manual implementation, the resulting coefficients were compared against those obtained using the standard `scikit-learn` linear regression implementation in Python.

2) *Multivariable Linear Regression*: Following validation in the univariate setting, the analysis was extended to multiple linear regression to model the combined influence of all predictors on calorie expenditure:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Linear regression was implemented using `statsmodels` and evaluated via the F -statistic, R^2 , and p -values, with residual normality verified using Q-Q plots.

I. Software Packages Used

The analysis was implemented in Python using the following packages:

TABLE I: Software Packages and Their Purposes

Package	Purpose
pandas (1.5.3)	Data manipulation, cleaning, and descriptive statistics
numpy (1.24.3)	Numerical computations and mathematical operations
scipy (1.10.1)	Statistical functions including IQR calculations
matplotlib (3.7.1)	Data visualization and plotting
seaborn (0.12.2)	Statistical visualization and distribution plots
scikit-learn (1.2.2)	Machine learning algorithms and preprocessing

III. RESULTS AND DISCUSSION

A. Data Quality and Outlier Management

The dataset underwent rigorous quality assessment. Initial inspection of the 15,000 raw observations found no structural errors: no missing values, duplicates, or physically impossible

entries (e.g., negative duration or zero heart rate). Univariate outliers were identified using the Interquartile Range (IQR) method, detecting 385 observations (2.57%) outside the nominal range, which were removed to avoid distortion of least-squares regression estimates. Figure 1 shows feature distributions after cleaning, leaving 14,615 observations, satisfying the sample size requirement ($n \geq p + 5$) for multivariate regression.

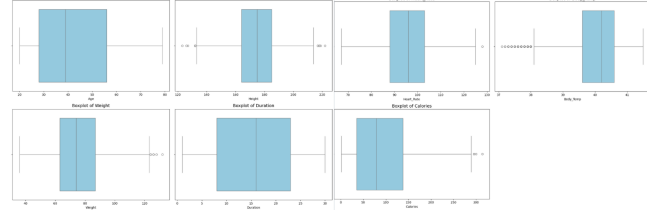


Fig. 1: Boxplots of all quantitative features following outlier removal, demonstrating the physiological range of the processed data.

B. Data Transformation Verification

To ensure the numerical stability of the regression model, feature standardization and categorical encoding were verified.

1) *Standardization Checks*: As shown in Table ??, the Z-score normalization successfully centered all continuous predictors. The means are effectively zero (0.0000) and standard deviations are exactly 1.0000. This confirms that features with large magnitudes (e.g., Height in cm) will not disproportionately bias the model coefficients compared to features with smaller units (e.g., Body Temp in °C).

2) *Categorical Encoding*: The one-hot encoding of the Gender variable produced a highly balanced binary distribution:

- **Females (0)**: 7,358 (50.3%)
- **Males (1)**: 7,257 (49.7%)

This near-even split reduces the risk of class imbalance bias, ensuring the model learns equally from both demographic groups.

C. Descriptive Statistics

Table ?? and Table ?? present the central tendency and dispersion metrics of the cleaned data.

Analysis of Central Tendency: The mean Age of 42.4 years suggests a middle-aged cohort, while the average Duration (15.5 min) and Heart Rate (95.5 bpm) indicate that the dataset primarily captures moderate-intensity exercise sessions. The difference between the Mean Calories (89.54) and Median (83.0) hints at a right-skewed distribution, suggesting that a small number of high-intensity sessions pull the average up.

Analysis of Variability: *Calories* exhibits the highest variance (3901.21), confirming it as the most volatile variable and a suitable target for prediction. Conversely, *Body_Temp* shows minimal deviation ($\sigma = 0.67$), reflecting strict biological homeostasis. The consistent spread in Height and Weight confirms the sample covers a diverse range of body types.

TABLE II: Summary of Central Tendency and Dispersion Measures

Variable	Mean	Median	Mode	Std Dev	Variance	Range	IQR
Age (years)	42.41	43	45	16.28	264.95	59	25
Height (cm)	175.96	176	176	14.01	196.28	99	21
Weight (kg)	74.98	74	70	15.98	255.36	96	24
Duration (min)	15.52	16	16	8.32	69.22	29	12
Heart Rate (bpm)	95.52	94	94	9.58	91.78	61	15
Body Temp (°C)	40.02	40.1	40.2	0.67	0.45	4.4	1
Calories (kcal)	89.54	83	66	62.46	3901.21	313	103

D. Statistical Distribution Analysis

Feature distributions were assessed via histograms, Q-Q plots, and the Anderson–Darling test; despite statistical non-normality from large sample size, visual inspection confirms practical usability.

1) *Feature-Specific Analysis*): Height (Anderson–Darling 18.92) is approximately Gaussian and symmetric, with normality formally rejected due to large sample size but practically well-conditioned. Weight (62.34) is roughly normal with minor peak irregularities, suitable for linear modeling. Heart Rate (27.92) is highly symmetric with strong Q-Q alignment, suggesting good normality. Body Temperature (228.89) is negatively skewed but manageable. Age (232.47) is unimodal and right-skewed with S-shaped Q-Q deviations, yet linear models remain robust. Duration exhibits a symmetric, platykurtic distribution, suitable for regression, while Calories (175.43), the target variable, is right-skewed with no transformation required as regression assumptions apply to residuals.

E. Data Transformation Outcomes

1) *Feature Standardization and Encoding*: Standardized features in the training data have mean ≈ 0.00 and standard deviation 1.00, ensuring numerical stability in OLS regression.

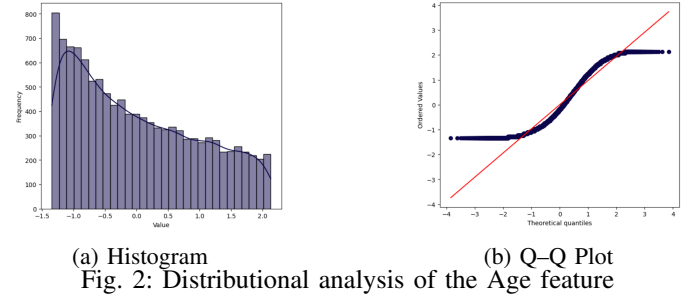
Gender was encoded via $k - 1$ One-Hot Encoding as Gender_male, with 7,358 females (0) and 7,257 males (1), showing near balance.

The dataset was split into training and test sets using stratified sampling on Gender, preserving proportions for fair evaluation.

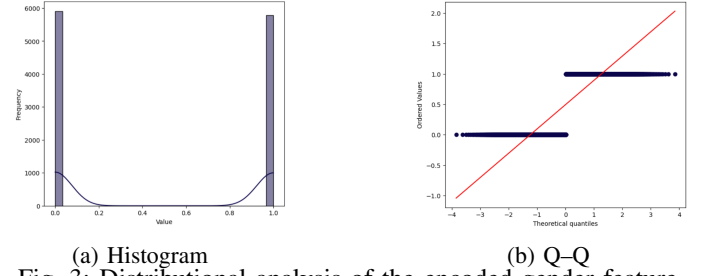
F. Statistical Analysis Results

This section summarizes the distributions of the training dataset’s predictors, combining visual diagnostics (histograms, Q-Q plots) with formal normality tests to assess distributional assumptions.

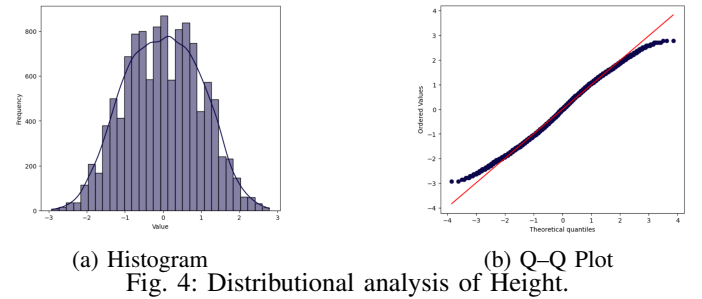
1) *Feature Analysis: Age*: The Age variable shows a right-skewed, non-Gaussian distribution, with Q-Q plot deviations confirmed by the Anderson–Darling test ($232.47 > 0.787$, $p < 0.05$). Despite this, the large sample size supports its inclusion in the linear regression model without transformation.



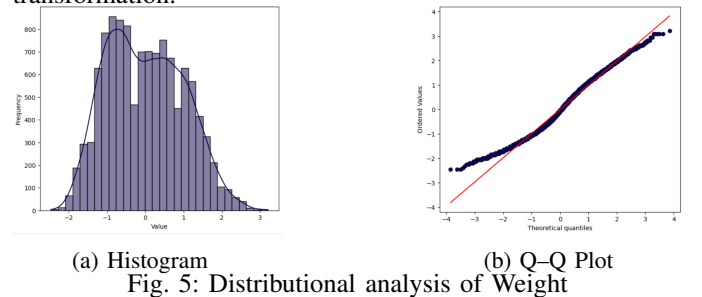
2) *Feature Analysis: Gender (Encoded)*: The Gender_male feature follows a binary Bernoulli distribution, with a step-pattern Q-Q plot. Normality is formally rejected (Anderson–Darling 2100.54), but class balance is excellent, with 7,358 Females and 7,257 Males, ensuring the regression model remains unbiased without transformation.



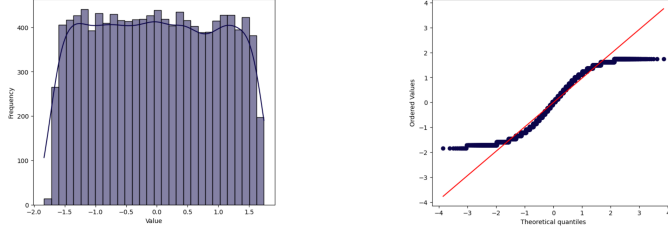
3) *Feature Analysis: Height*: The Height feature is symmetric and approximately Gaussian (Fig. 4a), with strong linear alignment in the Q-Q plot and only minor tail deviations (Fig. 4b). Normality was formally rejected (Anderson–Darling 18.92 > 0.787 ; $p < 0.05$) due to large sample size. In practice, Height is well-behaved and suitable for linear regression without transformation. .



4) *Feature Analysis: Weight*: The Weight feature is approximately symmetric with mild Q-Q plot tail deviations. Normality was rejected (Anderson–Darling 62.34 > 0.787 ; $p < 0.05$), largely due to sample size; however, Weight remains well-behaved and suitable for linear regression without transformation.

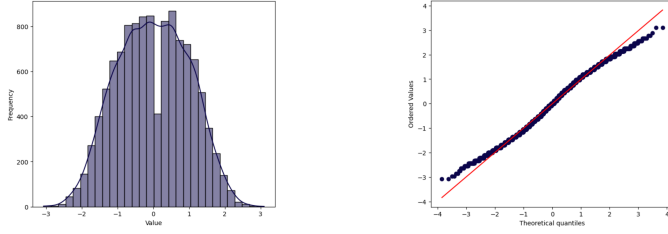


5) *Feature Analysis: Duration*: The `Duration` feature shows a flat-topped, symmetric (platykurtic) distribution (Fig. 6a, 6b). Normality is rejected (Anderson–Darling $132.22 > 0.787$; D’Agostino–Pearson $p \approx 0$), but the feature is well-bounded and informative, making it suitable for linear regression.



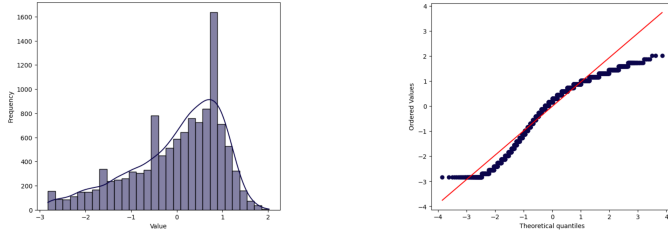
(a) Histogram (b) Q-Q Plot
Fig. 6: Distributional analysis of Duration.

6) *Feature Analysis: Heart Rate*: `Heart_Rate` exhibits a highly symmetric, near-Gaussian distribution (Fig. 7a) with strong Q-Q plot alignment (Fig. 7b). Despite formal rejection of normality (Anderson–Darling 27.92), the deviations are minor, rendering the variable suitable for linear regression without transformation.



(a) Histogram (b) Q-Q Plot
Fig. 7: Distributional analysis of Heart Rate.

7) *Feature Analysis: Body Temperature*: `Body_Temp` displays a left-skewed distribution (Fig. 8a) with lower-tail Q-Q plot curvature (Fig. 8b), reflecting physiological limits. Although normality was formally rejected (Anderson–Darling 228.89), the variable remains valid for regression due to the large sample size.



(a) Histogram (b) Q-Q Plot
Fig. 8: Distributional analysis of Body Temperature.

The `Calories` variable exhibits a right-skewed distribution with clear Q-Q plot deviations from normality. Formal testing rejected normality (Anderson–Darling $175.43 > 0.787$; $p = 3.41 \times 10^{-239}$); however, since regression requires normality of residuals rather than the response variable, and given the large sample size, no transformation was applied.

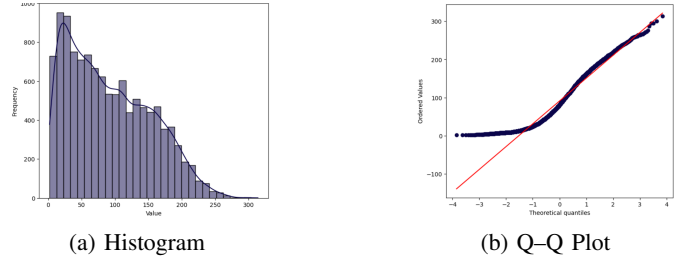


Fig. 9: Distributional analysis of Calories.

G. Correlation Analysis

To quantify the linear relationship between predictors and the response (`Calories`), the Pearson Correlation Coefficient (r) was computed. As defined in the statistical framework, r is calculated as the standardized covariance:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (2)$$

Results and Interpretation: Figure 10 visualizes the correlation matrix, while Table III details the coefficients.

TABLE III: Pearson Correlation Coefficients with Calories

Feature	Correlation (r)	Interpretation
Duration	0.955	Very Strong Positive
Heart_Rate	0.897	Very Strong Positive
Body_Temp	0.824	Strong Positive
Weight	0.583	Moderate Positive
Height	0.448	Moderate Positive
Age	0.154	Weak Positive

The analysis yields critical insights for model selection:

- 1) **Primary Predictors**: `Duration` ($r = 0.96$) and `Heart_Rate` ($r = 0.90$) exhibit near-linear dependencies, confirming that energy output is a direct function of intensity and time.
- 2) **Multicollinearity**: High correlations among predictors themselves suggest multicollinearity, which must be considered during coefficient interpretation.
- 3) **Weak Predictors**: `Age` ($r = 0.15$) is a secondary factor compared to physical effort metrics.

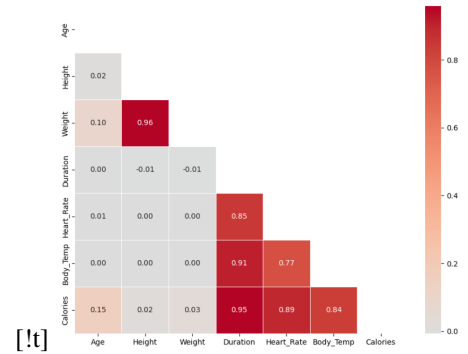


Fig. 10: Correlation matrix heatmap.

H. Regression Analysis

Comparing the manual implementation with `scikit-learn` (Table IV) shows identical slope (b_1) and intercept (b_0) values for all predictors, matching to five decimal places, confirming the mathematical accuracy of the implemented solution.

TABLE IV: Validation of Manual Regression Algorithm vs. Scikit-Learn (Univariate Coefficients)

Feature	Slope (b_1)		Intercept (b_0)	
	Manual	Sklearn	Manual	Sklearn
Age	9.38039	9.38039	91.09535	91.09535
Height	0.98277	0.98277	91.11262	91.11262
Weight	2.04542	2.04542	91.11347	91.11347
Duration	58.65577	58.65577	91.61174	91.61174
Heart Rate	55.04457	55.04457	91.48042	91.48042
Body Temp	51.23276	51.23276	91.53177	91.53177
Gender (Male)	2.65446	2.65446	89.79746	89.79746

The intercept b_0 is consistently around 91.1 to 91.6 for almost all features. Because we normalized the features only not the target, the intercept represents the average number of calories burned.

I. Analysis of Slope Coefficients and Model Performance

An analysis of the slope coefficients (β_1) shows the individual impact of each physiological feature:

Duration is the strongest predictor ($\beta_1 \approx 58.7$), where a one-standard deviation increase yields approximately 58.7 additional calories. **Heart Rate** is the second strongest predictor, with higher rates substantially increasing calorie expenditure. **Body Temperature** shows a strong positive univariate association, consistent with 60–70% of metabolic energy being released as heat. **Age** has a moderate positive effect, indicating slightly higher calorie burn with increasing age. **Gender** shows a small positive effect for males relative to other predictors. **Height** and **Weight** have minimal predictive value when considered without exercise intensity or duration.

1) *Multivariable Model Performance*: The regression quality was assessed at two levels: the overall model and individual coefficients.

- **Global Significance (F-Test)**: The model is statistically significant. The F-statistic (4.821×10^4) has a p -value of 0, leading to rejection of the null hypothesis that all coefficients are zero.
- **Goodness of Fit (R^2)**: The model explains 96.7% of the variance in calorie burn ($R^2 = 0.967$), with Adjusted $R^2 = 0.967$, indicating minimal risk of overfitting.

2) *Coefficient Analysis and Multicollinearity*: To assess the contribution of each predictor individually, we analyzed the regression coefficients (β), standard errors, and the results of the t-tests (t statistic and $P > |t|$).

Multicollinearity Assessment: Comparison of univariate and multivariable results reveals multicollinearity, as the Body Temperature coefficient changes from positive (51.23) to negative (−12.01). This sign reversal reflects shared variance with Duration and Heart Rate, adjusting the heat effect once exercise intensity is accounted for.

TABLE V: Multivariable Regression Coefficients (OLS Results)

Variable	Coef	Std Err	t	P > t
Const	92.2875	0.202	456.328	0.000
Age	8.6246	0.110	78.543	0.000
Height	-2.8785	0.392	-7.341	0.000
Weight	4.8020	0.448	10.720	0.000
Duration	53.7586	0.302	178.210	0.000
Heart_Rate	18.8064	0.195	96.403	0.000
Body_Temp	-12.0129	0.252	-47.587	0.000
Gender_male	-1.4216	0.350	-4.057	0.000

3) *Residual Diagnostics*: The normality of residuals was assessed using a Q–Q plot (Fig. 11). Residuals are approximately normal in the center but show heavier-than-normal tails. Despite formal normality rejection, the high R^2 confirms strong model performance.

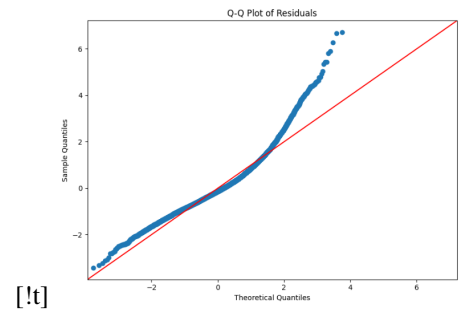


Fig. 11: Q–Q plot of residuals.

IV. CONCLUSION

This study established that linear regression accurately predicts calorie expenditure based on physiological and demographic inputs. Exercise duration and heart rate emerged as dominant predictors, while age and body dimensions exhibited minimal influence. The robust multivariable model achieved high explanatory power, validating linear regression as a reliable tool for modeling calorie burn in extensive exercise.

CONTRIBUTIONS

- **jana Gamal**: Data selection, preprocessing: split, cleaning, outlier removal, hot encoding and standarization.
- **Nardeen Ezz**: Feature analysis: Comprehensive Descriptive Statistics, histograms, normality tests (Anderson–Darling (AD), D’Agostino–Pearson (DP), Q-Q).
- **Yousef samy**: Simple linear regression: manual coefficient derivation, visualization, and package comparison (statsmodels, scikit-learn).
- **Farah Yehya**: feature correlation analysis and Multivariable regression model, organised all results for paper and presentation.

V. REFERENCES

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